

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 3.3 EXCRETION AND HOMEOSTASIS IN ANIMALS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions? Think back to the hot afternoon in a Kenyan community: a dog panting with its tongue hanging out, a lizard sitting still on a warm rock, goats that drink water yet make concentrated urine, and freshwater fish that keep excreting large amounts of dilute urine. Answer each section in the blank space under the prompt. Use evidence from what you observed in Lessons 1 to 6 (charts, the goat or cow kidney dissection, the nephron animations, the liver and skin photomicrographs, the feedback flowcharts and the thermoregulation notes). Write in full sentences and use correct biology words. Where a section asks for a claim, evidence and reasoning, give all three. At the end you will use your answers to help build your annotated poster model of how a mammal regulates its internal environment and excretes waste, which your teacher will mark with a rubric.

Section 1: Getting rid of wastes in different habitats

Insects, birds, freshwater fish and mammals such as goats make different nitrogenous wastes and use different excretory structures. (a) Name the nitrogenous wastes ammonia, urea and uric acid and state which animals in our charts excrete each one. (b) Name the excretory structures used by insects, birds, fish and mammals. (c) Explain how the type of waste an animal makes is linked to how much water is available to it. Use the freshwater fish and the bird or insect as your two examples.

(a) Ammonia is excreted by freshwater fish, which have plenty of water. Urea is made by mammals such as goats and humans and by marine cartilaginous fish such as sharks. Uric acid is excreted by insects and birds as a semi-solid paste. (b) Insects use Malpighian tubules; birds and mammals use kidneys; fish use gills and kidneys. (c) Ammonia is very toxic and must be diluted with a lot of water, so it suits freshwater fish, which are surrounded by water and gain it by osmosis and so can excrete large amounts of dilute urine. Birds and insects must save water, so they make uric acid, which is much less toxic and can be removed as a semi-solid paste with very little water. Mammals make urea, which is less toxic than ammonia and can be carried in the blood and removed in urine. So the waste type matches the water available in the habitat, which allows each animal to remove toxic nitrogen without harming its cells.

Section 2: From blood to urine in the goat

On the hot afternoon the goat

(a) The cortex is the dark outer region containing many nephrons, where blood is filtered. The medulla is the paler, striped region

drank water yet produced concentrated urine. Using what you learned from the goat or cow kidney dissection and the nephron animations: (a) Describe the roles of the cortex, medulla and pelvis. (b) Explain, in order, how ultrafiltration, selective reabsorption and secretion in the nephron turn blood into urine. (c) Explain why glucose is absent from urine but urea is concentrated in it.	containing parts of the nephrons and collecting ducts, which help concentrate urine. The pelvis is the hollow space that collects urine and passes it to the ureter. (b) Blood enters the glomerulus under high pressure, and small molecules (water, glucose, urea, salts) are forced into Bowman's capsule; this is ultrafiltration. Blood cells and proteins are too big and stay in the blood. As the filtrate flows along the tubule, useful substances are taken back into the blood by selective reabsorption, and extra wastes and ions are added by secretion. Water is reabsorbed in the loop of Henle and collecting duct, so the urine becomes concentrated and passes to the pelvis and ureter. (c) All the glucose is reabsorbed in the proximal tubule because the body needs it, so none remains in urine. Urea is a waste and most of it stays in the filtrate, while water is taken back into the blood, so urea becomes more concentrated in the urine.
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Section 3: Liver and skin as partners of the kidney

Urea is found in the goat's urine and a little urea is found in sweat. (a) Explain where the urea comes from and how it reaches the kidney. (b) State two other jobs of the liver and what the skin excretes. (c) A resource person spoke about cirrhosis, hepatitis and fatty liver disease. Explain why damage to the liver would be a problem for the whole body.	(a) Urea is made in the liver when excess amino acids are deaminated. It is carried in the blood to the kidney, where it is filtered and removed in urine. (b) The liver breaks down haemoglobin to make bile pigments and processes toxic substances. The skin excretes sweat containing water, salts and a little urea, and the evaporation of sweat also helps cool the body. (c) If the liver is damaged by cirrhosis, hepatitis or fatty liver disease, it cannot deaminate amino acids, make bile pigments or process toxins properly. Toxic substances would build up in the blood and harm cells throughout the body, and the kidney and skin would not be able to make up for the lost liver work. This shows that the liver, kidney and skin work together to remove wastes and keep the body stable.
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Section 4: Negative feedback in water, glucose and temperature

(a) Explain what negative feedback means. (b) Describe the feedback loop that adjusts how much water the goat's kidney reabsorbs on a hot day, and explain why the freshwater fish produces dilute urine. (c) Describe how insulin and glucagon keep blood glucose steady after a meal of ugali and sukuma wiki. (d) Choose diabetes mellitus or diabetes	(a) Negative feedback detects a change from a set point and triggers a response that reverses the change, bringing the condition back to normal. (b) When the goat loses water on a hot day, more ADH is released, so the collecting ducts reabsorb more water. The goat makes a small amount of concentrated urine and saves water. Once water balance is restored, less ADH is released. The freshwater fish gains water by osmosis, so it excretes large amounts of dilute urine to get rid of the extra water. (c) After a meal, blood glucose rises and insulin is released, which lowers blood glucose. When blood glucose falls too low, glucagon is released, which raises it. Together they keep glucose within a narrow range. (d) In diabetes mellitus, the insulin/glucagon loop fails, so blood glucose is not brought down and sugar appears in the urine, with thirst and a lot of urine. (Or: in diabetes insipidus, the ADH loop fails, so the kidney does not reabsorb enough water and the person passes a lot of very dilute urine and feels very thirsty.)
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insipidus and explain which part of its loop has failed, using its symptoms.	
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Section 5: Final explanation of the hot afternoon	
Write a claim-evidence-reasoning explanation that answers the driving question using the dog, the lizard, the goat and the freshwater fish. Include: (a) how the dog and the lizard each keep a steady body temperature, and the difference between them; (b) how the goat and the freshwater fish handle water and wastes differently; (c) why excretion and homeostasis are essential for survival. Then state one idea from your Lesson 1 model that you have changed.	Claim: Animals face different environments, but each keeps its cells in a stable internal environment and removes toxic wastes using structures and behaviours suited to its habitat. Evidence and reasoning: (a) The dog is an endotherm. It pants, so water evaporating from its tongue and airways carries heat away, and its skin blood vessels dilate (vasodilation) to lose heat. Thermoreceptors detect the change, the hypothalamus is the control centre, and effectors reverse the change by negative feedback. The lizard is an ectotherm and relies mainly on behaviour: it basks to gain heat and moves into shade or a crevice when too hot, so it sits still on the warm rock. Both use homeostasis to survive heat. (b) The goat's nephrons filter blood and, with more ADH on a hot day, reabsorb water to make small amounts of concentrated urine that contains urea made in the liver. The freshwater fish gains water by osmosis and excretes large amounts of dilute urine with ammonia. (c) Without excretion, toxic wastes such as ammonia and urea would build up and damage cells. Without homeostasis, changes in temperature, water and blood glucose would stop cells working. Because each structure or behaviour suits the animal's habitat, its cells survive and it stays healthy. Change: at first I thought animals only differ in how they look; now I know the differences are adaptations for removing wastes and keeping the body stable.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Accuracy of excretion science (wastes, kidney, liver, skin)	Correctly names ammonia, urea and uric acid with the animals and structures, and accurately explains ultrafiltration, selective reabsorption, secretion and the roles of the liver and skin, linking waste type to water availability.	Mostly correct explanation of wastes, nephron steps and the roles of the liver and skin, with minor gaps or one weak link to water availability.	Several errors or missing parts; confuses wastes, structures or nephron steps, or lists facts without explaining.
Explanation of homeostasis and negative feedback	Clearly explains negative feedback and applies it correctly to ADH and water, insulin and glucagon and temperature control, including the role of the hypothalamus and effectors and the difference between the dog and the lizard.	Explains negative feedback and applies it to most examples with small omissions.	Describes changes without showing a loop, or confuses the responses of endotherms and ectotherms.
Use of evidence and claim-evidence-reasoning	Gives a clear claim supported by specific evidence from the lessons (dog, lizard, goat, freshwater fish, dissection, case	Gives a claim with some evidence from the lessons, but the reasoning is partly missing.	Claim is unclear or evidence is vague, missing or not from the lessons.

	studies) and reasoning that links evidence to the claim.		
Answering the driving question and reflecting on learning	Fully connects all four animals to the driving question, explains why excretion and homeostasis are essential for cell survival, and states a specific change from the Lesson 1 model.	Connects most animals to the driving question and gives a general reason and a simple change in thinking.	Leaves out animals or gives no reason why excretion and homeostasis matter; no clear reflection.
Scientific vocabulary and clarity	Uses scientific terms correctly and consistently in well-organised, clear sentences.	Uses most terms correctly; writing is mostly clear.	Uses few scientific terms or uses them incorrectly; writing is hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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